



Numerical investigations of assisted heating cold start strategies for proton exchange membrane fuel cell systems

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ABSTRACT

To investigate cold start strategies at the system level, an integrated transient system model is developed, consisting of stack, membrane humidifier, electrochemical hydrogen pump, compressor, and radiator. The unassisted startup from $-10\text{ }^{\circ}\text{C}$ succeeds while it fails when started from $-20\text{ }^{\circ}\text{C}$. To achieve the successful startup from $-20\text{ }^{\circ}\text{C}$, various assisted strategies are adopted. For reactant gas heating method, the additional heat carried by gases is averaged about 1.2 W for each individual cell when the temperature of humidifier and hydrogen pump are maintained at $60\text{ }^{\circ}\text{C}$. Meanwhile, a large amount of moisture is introduced to the stack, which may lead to accelerated failure. For stack heating method, the startup succeeds if the total heating power reaches 40 W . However, the corresponding temperature difference within stack reaches as large as $22.0\text{ }^{\circ}\text{C}$, which indicates that improving the thermal conductivity of fuel cell materials is of great importance. Under coolant heating method, the startup succeeds if the coolant temperature reaches $-5\text{ }^{\circ}\text{C}$, and the ice formation can even be avoided if the coolant temperature is kept at $10\text{ }^{\circ}\text{C}$. However, the power consumption for heating coolant is extremely large, indicating that secondary power sources are necessary.

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1. Introduction

Owing to the advantages of high power density, high energy conversion efficiency, and zero emissions, proton exchange membrane fuel cell (PEMFC) is considered as one of the most important power sources for the automobile transportation industry in the prospective hydrogen society [1–3]. Several giant automobile companies have already made fuel cell vehicles available to the public, such as Mirai from Toyota [4], Clarity from Honda [5], and Nexa from Hyundai [6]. Note that the powertrain technologies in Clarity and Nexa have been both rated as Wards ten best engines [7]. The application of hybrid fuel cell systems in unmanned aerial vehicle [8], forklift [9], and ship [10] has also been reported. However, there still remain some technical challenges before the large-scale commercialization, among which the successful and rapid startup from subzero temperatures is of great importance [11]. Based on the ultimate targets of 80-kW integrated

transportation fuel cell power systems proposed by the U.S. Department of Energy (DOE), unassisted startup from $-30\text{ }^{\circ}\text{C}$ and assisted startup from $-40\text{ }^{\circ}\text{C}$ are required [12]. Less than 30 s is required for the fuel cell power system to achieve 50% rated power when started from $-20\text{ }^{\circ}\text{C}$ [12]. As regards the startup/shutdown durability, the output voltage deterioration after 5000 cycles is required to be less than 5% under specific testing protocols [12]. Despite the fact that many cold start researches have been conducted, the aforementioned technical targets are still hard to be achieved, which requires further investigations [11].

During cold start processes, the generated water through hydrogen-oxygen reactions is likely to freeze if the local temperature is lower than the freezing point. Since ice is solid, it is not able to move out of the porous layers, thus blocking effective reaction sites in the catalyst layer (CL) and species transport pathways in the gas diffusion layer (GDL) [13]. Once the air flowing into the fuel cell is unable to reach three-phase boundaries (TPB) in the CL or the amount of oxygen that successfully reaches TPB is unable to compensate for the electrochemical reaction consumption, the startup is considered failed. From the perspective of thermal behavior, the cold start can be generalized as the competition

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Nomenclature		η	efficiency
A		<i>Subscripts and superscripts</i>	
ASR	effective geometric area (m^2)	a	anode
C_p	area specific resistance ($\Omega \text{ m}^2$)	act	activation
D	specific heat capacity ($\text{J kg}^{-1} \text{ K}^{-1}$)	atm	atmosphere
EW	diffusivity ($\text{m}^2 \text{ s}^{-1}$)	BP	bipolar plate
f	equivalent weight of membrane (kg kmol^{-1})	c	cathode
F	function	CL	catalyst layer
G	Faraday's constant (C mol^{-1})	cm	compressor motor
h	Gibbs free energy (J mol^{-1})	conc	concentration
I	heat transfer coefficient ($\text{W m}^{-2} \text{ K}^{-1}$)	cond	condensation
J	current density (A m^{-2})	cool	coolant
	ionic current density (A m^{-2}), inertia of compressor	cp	compressor
	(kg m^2)	eff	effective
k	thermal conductivity ($\text{W m}^{-1} \text{ K}^{-1}$)	eq	equilibrium
K	permeability (m^2)	FC	fuel cell
m	mass flow rate (kg s^{-1})	g	gas phase
M	molecular weight (g mol^{-1})	GDL	gas diffusion layer
p	pressure (Pa)	H_2	hydrogen
P	power (W)	HP	hydrogen pump
Q	heat source term (W m^{-3})	in	inlet
R	universal gas constant ($\text{J mol}^{-1} \text{ K}^{-1}$), motor armature	ion	ionic
	resistance (Ω)	l	liquid phase
s	volume fraction	lq	liquid water
S	source terms ($\text{kmol m}^{-3} \text{ s}^{-1}$, $\text{kg m}^{-3} \text{ s}^{-1}$, W m^{-3}),	mem	membrane
	entropy ($\text{J mol}^{-1} \text{ K}^{-1}$)	MH	membrane humidifier
ST	stoichiometry	mw	membrane water
t	time (s)	MPL	micro-porous layer
T	temperature (K)	O_2	oxygen
u	velocity (m s^{-1})	ohmic	ohmic
V	voltage (V)	out	output
Y	mass fraction of gas species	pc	phase change
Greek letters		per	permeation
ε	porosity	react	reaction
λ	membrane water content	ref	reference state
μ	dynamic viscosity ($\text{kg m}^{-1} \text{ s}^{-1}$)	req	required
ρ	density (kg m^{-3})	sat	saturation
ω	volume fraction of ionomer, compressor rotation	surr	surroundings
	angular frequency (rad s^{-1})	vp	water vapor
δ	thickness (m)	m-l	membrane water to super-cooled/liquid water
κ	motor torque constant	m-v	membrane water to water vapor
τ	torque (N m)	l-i	super-cooled water to ice
v	motor terminal voltage (V)	v-l	water vapor to liquid water

between ice formation and temperature increment [14]. If the cell temperature rises above the freezing point in ahead of the time when catalytic reaction sites or gas transport passages are fully blocked by ice, the startup is successful. Since the volume expands when liquid water is turned into ice, irreversible damage may be caused to microstructures inside the fuel cell during freezing/thawing cycles, including pinhole or crack of membrane [15,16], agglomeration of catalyst particles [17,18], delamination between CL and GDL [19,20], and other detrimental influences [11,21,22]. To investigate the ice formation mechanisms and corresponding impacts on fuel cell performances, both experimental and numerical researches have been conducted. Effects of operating and structural parameters (such as startup temperature [15], current density [23,24], membrane dehydration level [25,26], reactant gas flow arrangement [27], flow-field configuration [28,29], and ionomer/carbon ratio [30]) were the focus of experimental studies. To further characterize local current density and temperature

distribution, the segmented fuel cell technology [27,31] and the printed circuit board technology [29,32] have been adopted. With visualization technologies been greatly developed, more researches were aimed at directly demonstrating ice generation processes inside porous layers and flow channels based on methodologies such as transparent fuel cell [33], infrared image [34], Raman spectroscopic examination [35], and scanning electron microscope [36–38]. In addition to experimental studies, various simulation models have been developed to explain the coupled heat and mass transport phenomena during sub-zero temperatures, including lumped thermal model [39], analytical model [40], one-dimensional [41–44], two-dimensional [45,46], and three-dimensional models [47–49]. The application of artificial intelligence for numerical models [50–52] and optimization algorithms for model parameter estimation [53,54] have also been presented. Owing to worldwide researches over the past two decades, ice formation mechanisms have been well revealed, and a large

number of cold start strategies have been consequently proposed. From the perspective of fuel cell structural designs, the addition of micro-porous layer was widely considered to be beneficial [55–57]. As for preconditioning methods, the gas purge process after shut-down was necessary because it minimized the amount of residual water [25,26,58]. For startup operating conditions, various startup modes and strategies were proposed such as maximum power startup mode [59], variable heating and load control [60], and model-based real-time adaptive control [61]. Besides, adopting the methanol-water solution as antifreeze agent was found to inhibit performance degradation after freezing/thawing cycles [62]. Note that the above strategies were focused on the self-startup of single cell or stack, requiring no additional heating power consumption but relatively long warm-up time. Consequently, assisted cold start strategies such as reactant gas heating [63], stack heating [64], coolant heating [65], hydrogen-oxygen catalytic reaction [66,67], and external hydrogen burner [68] were proposed to accelerate the startup process. However, a comprehensive and quantitative comparison among those assisted startup strategies, which considered the operability, timeliness, efficiency, power consumption, and local parameter distribution inside individual fuel cells (such as current density, ice, temperature), was few to find in literature. Besides, the majority of aforementioned cold start researches were concentrated on characterizing and optimizing the startup capacity of single fuel cells or stacks, which hardly addressed the problem at a comprehensive PEMFC system level. As presented previously [69,70], a typical PEMFC system consisted of stack, reactant gas supply subsystems, humidification subsystem, and heat management subsystem. The hydrogen supply subsystem consists of high-pressure hydrogen tank, pressure-regulating valve, three-way valve, water knockout, and hydrogen recirculation pump. The air supply subsystem includes air filter, compressor, radiator, membrane humidifier, water knockout, and solenoid valve. The heat management subsystem is composed of water tank, pump, heater, thermostat, ion collector, and radiator. The stable operation of system was jointly guaranteed by the coordination among all subsystems while the couple water and heat management within the whole system was extremely sophisticated [69,71]. Take the water transport for an example, the humidification of membrane humidifier relied on the stack cathode exhausted gases while it had great influences on stack performances during cold start processes. The successful systematic startup from subzero temperatures should not only raise the stack temperature over the freezing point rapidly but also simultaneously accelerate the temperature increment of auxiliary components without sufficient internal heat generation (such as membrane humidifier). However, few related researches were presented in literature, making it necessary to investigate the cold start issue from the view of an authentic system level.

In the study, an integrated transient PEMFC system model that has been developed in our previous studies [69,70] is adopted to investigate the cold start capability of whole system. The unassisted systematic cold start performances under different startup operating conditions are briefly investigated, including the distribution of output voltage, ice volume fraction, and temperature. To achieve the successful startup from subzero temperatures as low as $-20\text{ }^{\circ}\text{C}$, assisted startup strategies such as reactant gas heating method, stack heating method, and coolant heating method are quantitatively compared. The output performance, stack inlet reactant relative humidity, ice generation rate, temperature distribution, and cold start duration are comprehensively investigated. The effectiveness of assisted startup strategies is evaluated, which gives valuable suggestions on cold start issues.

2. Model development

Methodologies about the core sub-models are briefly presented, including stack, membrane humidifier, electrochemical hydrogen pump, compressor, and radiator. The integration of various cold start strategies and the system model is also presented.

2.1. Stack model

Proton exchange membrane fuel cell (PEMFC) typically consists of bipolar plate (BP), gas channel, gas diffusion layer (GDL), micro-porous layer (MPL), catalyst layer (CL), and proton exchange membrane. For the stack composed of many single cells, the amount of reactant gases and coolant flowing into each individual fuel cell are typically different owing to pressure losses during the motion of fluids [72,73]. However, it is found that the flow distribution uniformity can be significantly improved through the optimization of stack manifold geometry in our recent study [74]. Besides, low current densities (such as 0.1 A cm^{-2}) are widely adopted in cold start operations, which reduces the distribution non-uniformity to a large extent because the mass flow rate of reactants is relatively small. Therefore, it is reasonable to assume that reactant gases and coolant are uniformly distributed into every fuel cell in the presented study. In addition, mass transport processes in porous layers are assumed to be diffusion-dominated in the through-plane direction [75,76]. The product water is assumed to be membrane water initially according to the previous study [44] in which the phase transition mechanisms during cold start processes have been comprehensively investigated. Phase changes among water vapor, membrane water, super-cooled/liquid water, and ice are fully taken into consideration. Heat radiation between the stack and surroundings is neglected while heat convection is taken into account. The stack structural properties and operating conditions are listed in Table 1. The governing equations of membrane water, super-cooled/liquid water, gas species, ice, temperature, and output voltage are summarized in Table 2 while the source terms are summarized in Table 3. Note that the heat and mass transfer processes are calculated in the through-plane direction as well as the gas channel direction. Therefore, the parameter variations from upstream to downstream can be clearly demonstrated, including local current density, ice volume fraction, temperature, and other key parameters [69,70]. Besides, performance differences (such as output voltage and temperature) among various single fuel cells can also be well demonstrated. The detailed correlations for electrochemical and transport properties have been presented in the previous studies [43,69].

During cold start processes, it should be mentioned that the authentic freezing temperature in porous layers is lower than the normal freezing point ($T_N = 273.15\text{ K}$), which is called the freezing point depression [13,43]. The temperature difference is defined as:

$$T_{\text{FPD}} = \frac{T_N \sigma_{273.15} \cos(\theta)}{\rho_{\text{ice}} h_{\text{fusn}} r_{\text{CL,MPL,GDL}}} \quad (1)$$

where $\sigma(\text{N m}^{-1})$ is the surface tension, $\theta(^{\circ})$ is the contact angle, $h_{\text{fusn}}(\text{J kg}^{-1})$ is the latent heat of fusion, $r(\text{m})$ is the pore radius of porous layers.

Special attention should be paid to the heat source term since assisted startup heating sources are embedded. The amount of heat carried by hot inlet reactants is represented by Q_{gas} , and the heat absorbed from coolant is represented by Q_{coolant} . For the stack heating method, a specific heating source is added to the end plate at two sides of stack. More detailed information about the pseudo two-dimensional stack model can be found in our previous studies [69,70].

Table 1
Stack structural properties and operating conditions.

Parameter	Value
Number of fuel cells	10
Effective area of a single fuel cell	200 cm ²
Effective heat transfer area between coolant and a single fuel cell	100 cm ²
Channel length, width, depth, rib width	200; 1.0; 1.0; 1.0 mm
Thickness of membrane, CL, MPL, GDL, BP	0.050; 0.01; 0.03; 0.2; 2 mm
Density of membrane, CL, MPL, GDL, BP	1980, 1000, 1000, 1000, 1000 kg m ⁻³
Specific heat capacity of membrane, CL, MPL, GDL, BP	833, 3300, 568, 2000, 1580 J kg ⁻¹ K ⁻¹
Thermal conductivity of membrane, CL, MPL, GDL, BP	0.95, 1.0, 1.0, 1.0, 20 W m ⁻¹ K ⁻¹
Electric conductivity of membrane, CL, GDL, BP	5000, 5000, 5000, 20,000 S m ⁻¹
Ionomer volume fraction in CL	0.4
Contact angle of CL, MPL, GDL	100°, 120°, 120°
Porosity of CL, MPL, GDL	0.4, 0.5, 0.7
Requested stoichiometry ratio	ST _a ^{req} = 1.2, ST _c ^{req} = 2.0
Requested operating pressure	p _{a,c} ^{req} = 1.1 atm
Initial membrane water content	3.2
Initial super-cooled water and ice volume fraction	0
Initial stack and surrounding temperature	253, 263 K
Heat transfer coefficient between endplate and surrounding	2 W m ⁻² K ⁻¹
Total coolant mass flow rate	0.1 kg s ⁻¹
Heat transfer coefficient between coolant and a single fuel cell	200 W m ⁻² K ⁻¹

Table 2
Governing equations of the stack model.

Parameter	Equation
Membrane water	$\frac{\rho_{mem}}{EW} \frac{\partial(\omega\lambda)}{\partial t} + \nabla \cdot \left(n_d \frac{J_{ion}}{F} \right) = \frac{\rho_{mem}}{EW} \nabla \cdot (\omega^{1.5} D_{mw} \nabla \lambda) + S_{mw}$
Super-cooled/Liquid water	$\begin{cases} \frac{\partial(\epsilon \rho_{lq} S_{lq})}{\partial t} = \nabla \cdot \left(\rho_{lq} \frac{K_{lq}}{\mu_{lq}} \nabla p_l \right) + S_{lq} & (\text{In CL, MPL, GDL}) \\ \frac{\partial(\rho_{lq} S_{lq})}{\partial t} + \nabla \cdot (\rho_{lq} u_{lq} S_{lq}) = \nabla \cdot \left(\rho_{lq} \frac{K_{lq}}{\mu_{lq}} \nabla p_l \right) + S_{lq} & (\text{In channel}) \end{cases}$
Ice	$\frac{\partial(\epsilon \rho_{ice} S_{ice})}{\partial t} = S_{ice} (\text{In CL, MPL, GDL, channel})$
Gas species	$\begin{cases} \frac{\partial}{\partial t} (e(1 - S_{lq} - S_{ice}) \rho_g Y_i) = \nabla \cdot (\rho_g D_i^{eff} \nabla Y_i) + S_i & (\text{In CL, MPL, GDL}) \\ \frac{\partial}{\partial t} ((1 - S_{lq} - S_{ice}) \rho_g Y_i) + \nabla \cdot (\rho_g u_g Y_i) = \nabla \cdot (\rho_g D_i^{eff} \nabla Y_i) + S_i & (\text{In channel}) \end{cases}$
Temperature	$\frac{\partial}{\partial t} ((\rho C_p)_{fl,sl}^{eff} T) = \nabla \cdot (k_{fl,sl}^{eff} \nabla T) + S_T$
Voltage	$\begin{aligned} V_{out} &= V_{Nernst} - V_{act} - V_{ohmic} - V_{conc} \\ V_{Nernst} &= \frac{\Delta G}{2F} + \frac{\Delta S}{2F} (T - T_{ref}) + \frac{RT}{2F} \left[\ln \left(\frac{c_{H_2, CLa}}{c_{H_2, ref}} \right) + \frac{1}{2} \ln \left(\frac{c_{O_2, CLc}}{c_{O_2, ref}} \right) \right] \end{aligned}$

2.2. Membrane humidifier model

To utilize stack cathode exhausted gases, a planar membrane humidifier is adopted, which absorbs the moisture and sequentially humidifies the dry air. The membrane humidifier model mainly consists of gas channels and membrane. The addition of obstacles in flow field and GDL to enhance the diffusion of water vapor transfer has also been reported [77,78]. The ice formation inside membrane humidifier is neglected because the amount of super-cooled water along with stack cathode exhausted gases is tiny. It should be mentioned that there is no sufficient internal heat generation inside the membrane humidifier, and the operating temperature is likely to be much lower than that of stack [69]. Under the circumstances, water vapor condensation is possible to happen in the humidifier wet channel, but the quantity is minor, which has negligible influences on humidifier performances. Therefore, the amount of internally generated super-cooled water is also neglected in the cold start operation. Governing equations of membrane water, water vapor, and temperature as well as corresponding source terms are given in Table 4.

2.3. Electrochemical hydrogen pump model

A hydrogen recirculation pump is placed in the hydrogen supply subsystem to partially recycle the used hydrogen, enhancing fuel utilization and anode reactant humidification [79,80]. The structure of electrochemical hydrogen pump is similar to that of PEMFC, and the corresponding structural properties of electrochemical pump are set the same as that of PEMFC. Detailed information of working principles and transport properties has been presented in our previous study [69,70], and the governing equations are listed in Table 5.

2.4. Air compressor model

The centrifugal air compressor model consists of three parts, including the curve fitting of compressor characteristics, the dynamic response of motor, and the proportional-integral-derivative (PID) control strategy. Based on the characteristics of air compressor in literature [81], the air mass flow rate is calculated as the function of pressure ratio and rotation speed based on polynomial fitting methods. As shown in Eq. (2), the dynamic response

Table 3
Source terms.

Correlation	Unit
$S_{mw} = \begin{cases} S_{react} - S_{m-v} - S_{m-l} & (\text{In CLc}) \\ -S_{m-v} - S_{m-l} & (\text{In CLa}) \end{cases}$	$\text{kmol m}^{-3} \text{ s}^{-1}$
$S_{lq} = \begin{cases} S_{m-l}M_{H_2O} + S_{v-l} - S_{l-i} - S_{per} & (\text{In CLc}) \\ S_{m-l}M_{H_2O} + S_{v-l} - S_{l-i} + S_{per} & (\text{In CLa}) \\ S_{v-l} - S_{l-i} & (\text{In MPL, GDL, channel}) \end{cases}$	$\text{kg m}^{-3} \text{ s}^{-1}$
$S_{ice} = S_{l-i} (\text{In CL, MPL, GDL, channel})$	$\text{kg m}^{-3} \text{ s}^{-1}$
$S_{per} = \frac{\rho_{lq} K_{per} (p_{l,CLc} - p_{l,CLa})}{\mu_{lq} \delta_{CL} (0.5\delta_{CLa} + \delta_{mem} + 0.5\delta_{CLc})}$	$\text{kg m}^{-3} \text{ s}^{-1}$
$S_{vp} = \begin{cases} S_{m-v}M_{H_2O} - S_{v-l} & (\text{In CL}) \\ S_{v-l} & (\text{In MPL, GDL, channel}) \end{cases}$	$\text{kg m}^{-3} \text{ s}^{-1}$
$S_T = \begin{cases} \frac{l^2 ASR_{CL}}{3\delta_{CL}} + \frac{IV_{act,a}}{\delta_{CL}} + S_{pc} & (\text{In CLa}) \\ \frac{l^2 ASR_{CL}}{3\delta_{CL}} - \frac{T\Delta S}{2F} + \frac{IV_{act,c}}{\delta_{CL}} + S_{pc} & (\text{In CLc}) \\ \frac{l^2 ASR}{\delta} & (\text{In membrane}) \\ \frac{l^2 ASR}{\delta} + S_{pc} & (\text{In MPL, GDL}) \\ \frac{l^2 ASR}{\delta} + S_{pc} - Q_{gas} & (\text{In channel}) \\ \frac{l^2 ASR}{\delta} + Q_{coolant} & (\text{In BP}) \end{cases}$	W m^{-3}
$Q_{gas} = \frac{A_{in} \sum_i \{c_p u_{in} c_{in} M (T_{channel} - T_{in})\}_i}{A_{cell} \delta}$ ($i = H_2, O_2, N_2, vp$)	W m^{-3}
$Q_{coolant} = h_{cool} A_{cool} (T_{coolant}^{ave} - T_{BP})$	W m^{-3}

Table 4
Governing equations of the membrane humidifier model.

Parameter	Equation
Membrane water	$\frac{\rho_{mem}}{EW} \frac{\partial(\lambda_{MH})}{\partial t} = \frac{\rho_{mem}}{EW} \nabla \cdot (D_{MH,mw} \nabla \lambda_{MH}) + S_{MH,mw}, S_{MH,mw} = -S_{MH,m-v}$
Water vapor	$\frac{\partial}{\partial t} (\rho_g Y_{MH,vp}) + \nabla \cdot (\rho_g u_g Y_{MH,vp}) = S_{MH,vp}, S_{MH,vp} = S_{MH,m-v} M_{H_2O}$
Temperature	$\frac{\partial}{\partial t} (c_p^{eff} T_{MH}) = \nabla \cdot (k^{eff} \nabla T_{MH}) + S_{MH,T},$ $S_{MH,T} = \begin{cases} Q_{MH,gas} & (\text{In channel}) \\ h_{cond} (-S_{MH,m-v} M_{H_2O}) & (\text{In membrane}) \end{cases}$

Table 5
Governing equations of the electrochemical hydrogen pump model.

Parameter	Equation
Membrane water	$\frac{\rho_{mem}}{EW} \frac{\partial(\omega \lambda_{HP})}{\partial t} + \nabla \cdot \left(n_d \frac{J_{HP,ion}}{F} \right) = \frac{\rho_{mem}}{EW} \nabla \cdot (\omega^{1.5} D_{HP,mw} \nabla \lambda_{HP}) + S_{HP,mw}$
Gas species	$\begin{cases} \frac{\partial}{\partial t} (\epsilon \rho_g Y_{HP,i}) = \nabla \cdot (\rho_g D_{HP,i}^{eff} \nabla Y_i) + S_{HP,i} & (\text{In porous layers}) \\ \frac{\partial}{\partial t} (\rho_g Y_{HP,i}) + \nabla \cdot (\rho_g u_g Y_{HP,i}) = \nabla \cdot (\rho_g D_{HP,i}^{eff} \nabla Y_i) + S_{HP,i} & (\text{In channel}) \end{cases}$
Temperature	$\frac{\partial}{\partial t} (c_p^{eff} T_{HP}) = \nabla \cdot (k^{eff} \nabla T_{HP}) + S_{HP,T}$
Drive voltage	$V_{HP,Nernst} = \frac{RT}{2F} \ln \left(\frac{p_{HP,c}}{p_{HP,a}} \right), V_{HP,drive} = V_{HP,Nernst} + V_{HP,act} + V_{HP,ohmic}$

of compressor is decided by drive torque and load torque. Since the motor terminal voltage controls the drive torque, a PID control strategy is adopted to achieve the requested pressure ratio and mass flow rate. Detailed information about polynomial fitting

results, temperature and pressure calibration, and calculation procedure can be found in our previous studies [69,70].

$$\begin{cases} J_{cp} \frac{d\omega_{cp}}{dt} = \tau_{cm} - \tau_{cp} \\ \tau_{cm} = \eta_{cm} \frac{k_t}{R_{cm}} (v_{cm} - k_v \omega_{cm}) \\ \tau_{cp} = \frac{P_{cp}}{\omega_{cp}} \end{cases} \quad (2)$$

2.5. Radiator model

During the normal operation of stack, the radiator is supposed to dissipate the heat carried by coolant to surroundings. The ribbon-tubular fin radiator is used, and the heat transfer processes between coolant and surroundings can be divided into three stages, which includes heat convection between coolant and tube, heat conduction inside tube wall, and heat convection between tube and air. Under steady-state conditions, the released heat from coolant is equal to the absorbed heat by air.

$$Q = m_{lq}(c_p)(T_{lq}^{in} - T_{lq}^{out}) = m_{air}(c_p)_{air}(T_{lq}^{out} - T_{lq}^{in}) = AK\Delta t_m \quad (3)$$

$$\frac{1}{AK} = \frac{1}{h_{lq}A_{lq}} + \frac{\delta}{\lambda_t A_{lq}} + \frac{1}{h_{air}A_{air}\eta_0} \quad (4)$$

Based on the three different heat transfer stages, the overall heat transfer coefficient is calculated based on the specific radiator structures. The effectiveness-NTU method is adopted to calculate the outlet temperatures of coolant and air. Detailed calculation methods have been presented in our previous studies [69,70].

2.6. Boundary and initial conditions

The previous subsections have presented numerical methodologies of key sub-models, which are integrated as the comprehensive system model by coupling the boundary conditions as shown in Fig. 1. The hydrogen tank, regulating valves, water knockout

devices, coolant pump, and other less important components are neglected in the figure.

For the compressor model, the requested pressure ratio and air mass flow rate are determined by stack operating conditions. By regulating the PID-controlled motor terminal voltage, the real compressor rotating speed and pressure ratio are decided, thus determining the real air mass flow rate exiting the compressor.

$$m_{air}^{req} = \frac{N_{stack} I A_{cell} S T_c M_{air}}{4F \times 0.21 \times 1000} \quad (5)$$

The total amount of requested hydrogen is calculated according to Faraday's law while the stack anode inlet gases consist of the electrochemical hydrogen pump cathode outlet gases and the pure hydrogen from high-pressure tank.

$$m_{H_2}^{req} = \frac{N_{stack} I A_{cell} S T_a M_{H_2}}{2F \times 1000} \quad (6)$$

The total current of every fuel cell is identical owing to the connection in series. For a single fuel cell, the local current density varies from channel upstream towards downstream, and it is obtained by solving the system of nonlinear equations [69,70].

$$f(I_i) = V_{out}, \quad \sum_i^n I_i = n \cdot I \quad (7)$$

where n refers to the number of nodes in the gas channel direction.

As for the integration of assisted startup strategies with the system model, different methodologies are adopted. At the system level, it should be noted that the temperatures of reactants flowing into the stack are decided by auxiliary components. The stack cathode inlet temperature is determined by the membrane humidifier while the anode inlet temperature is calculated as the mixed temperature between recycled gases and pure hydrogen. Therefore, the reactant heating method is achieved by controlling the temperature of membrane humidifier and electrochemical hydrogen pump. The stack heating method is typically achieved by installing heating wires at the two sides of stack, thus directly raising the stack temperature by Joule heat. In the system model, a specific heat source is added to the two end plates. As for the

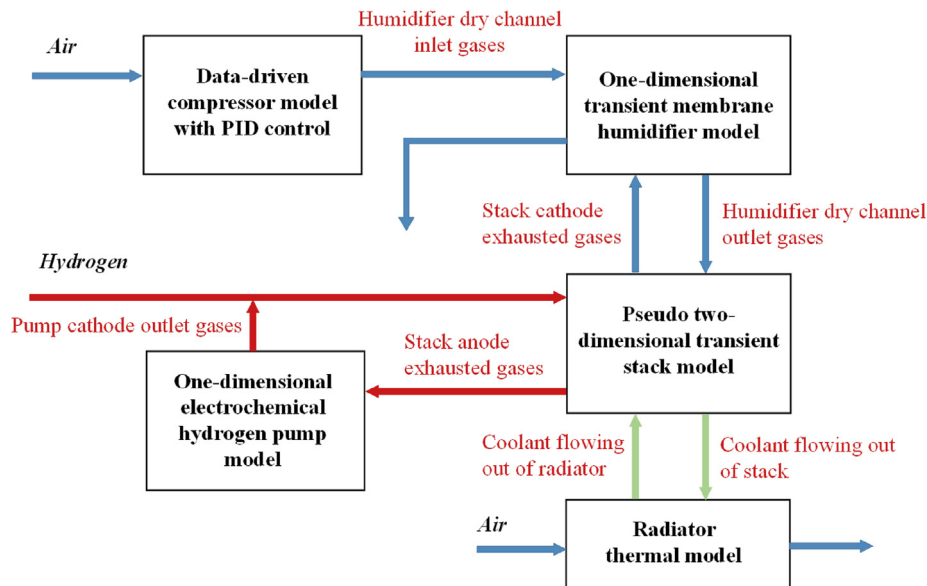


Fig. 1. Integration of all sub-models into the comprehensive system model.

coolant heating method, the coolant is heated until the expected temperature before entering the stack. The amount of heat absorbed by stack is treated as one of the heat source terms as listed in Table 3. Note that the radiator is turned off during cold start processes because there is no need for heat dissipation.

The initial temperature of components is set identical to the environment. The initial membrane water content of stack, membrane humidifier, and electrochemical pump is set as 3.2, indicating a relatively-long gas purge process after shutdown. The initial super-cooled water/ice volume fraction inside the stack is set as 0. As for numerical procedures, the explicit formulation calculation method is adopted for discretization with a time step size of 10^{-6} s. The parameters such as membrane water content, super-cooled water/ice volume fraction, and temperature are updated at the center of corresponding componential layers [69,70]. All models are established based on user defined functions in Matlab 2018a, and a workstation with 20 cores (2.4 GHz) is adopted to conduct numerical simulations.

3. Results and discussion

3.1. Model validation

To ensure the reliability of integrated proton exchange membrane fuel cell (PEMFC) system model, all sub-models have been validated against experimental data. For the stack sub-model, grid independence study, polarization curve at different cathode inlet relative humidity, and output voltage under failed/successful cold start conditions have been presented in the previous study [70]. In addition, the model has also been compared with our latest experimental data [82].

Fig. 2 shows the output voltage and the ohmic voltage when started from $-15\text{ }^{\circ}\text{C}$ and $-20\text{ }^{\circ}\text{C}$. The current density is set as 0.1 A cm^{-2} . Special attention should be paid to the beginning of corresponding experimental data in Ref. [82] because there is no current load in the first 10 s. Therefore, the actual and effective experimental data of startup processes is selected after the startup duration reaches 10 s, which is used to further validate the stack sub-model. Good agreement between our laboratory experiments and simulation results is observed. As for the four auxiliary sub-models, detailed model validation can also be found in the previous study [70].

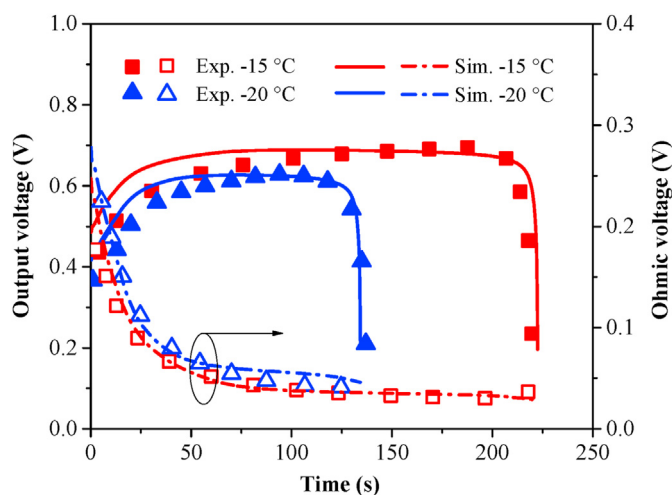


Fig. 2. Comparison of output voltage and ohmic voltage when started from $-15\text{ }^{\circ}\text{C}$ and $-20\text{ }^{\circ}\text{C}$ between experimental data [82] and simulation results.

3.2. Unassisted startup performances

Based on the previous study [70], the co-current flow configuration contributes to water vapor accumulation in the gas channel direction inside fuel cells, leading to higher water utilization of the overall system. Therefore, it is unadvisable to adopt the co-flow pattern since more ice is likely to be formed in the stack compared with the counter-current flow pattern. Consequently, the stack is placed in counter-flow configuration in the presented results. The unassisted startup performances are firstly investigated, which provide a benchmark against the following assisted startup strategies. The stack output voltage under the ambient temperature of $-10\text{ }^{\circ}\text{C}$ and $-20\text{ }^{\circ}\text{C}$ is shown in Fig. 3. The constant current density mode is adopted because it results in a relatively stable water generation rate, which contributes to the stack operation stability. The stack is successful to reach the freezing point temperature when started from $-10\text{ }^{\circ}\text{C}$ while it fails when started from $-20\text{ }^{\circ}\text{C}$. For the successful cases, a higher operating current density largely accelerates the startup speed owing to the higher amount of generated heat. In the failed cases, the startup duration is around 80 s at 0.1 A cm^{-2} while it only lasts for about 25 s with the current density of 0.2 A cm^{-2} . Since the stack is initially dry and the electro-osmotic drag effect becomes more significant with growing current densities, the membrane water content gradient between anode and cathode increases at 0.2 A cm^{-2} , leading to higher ohmic voltage losses and sharper output voltage drops in the beginning. Therefore, increasing the current density does not necessarily lead to enhanced cold start survivability because more water is simultaneously generated, which is likely to expedite the ice formation inside porous layers.

The distribution of ice volume fraction in cathode catalyst layer (CL) along the channel direction inside a single fuel cell is shown in Fig. 4(a), and the ice distribution among different fuel cells is shown in Fig. 4(b). Note that each fuel cell is divided into five nodes in the channel direction to reflect parameter variations from upstream to downstream, which is represented by the node number ranging from 1 to 5. The fuel cell 1 and 5 are particularly selected because they are positioned at the end and the middle of stack. In Fig. 4(a), there exist obvious ice volume fraction differences from the reactant inlet region to the outlet region. From node 2 to node 5, more ice is formed at the same startup time. The ice volume fraction in node 1 grows faster than that of node 2 and node 3. The main reason for above non-uniform ice formation rates is the variations of local current density, membrane water content, and temperature

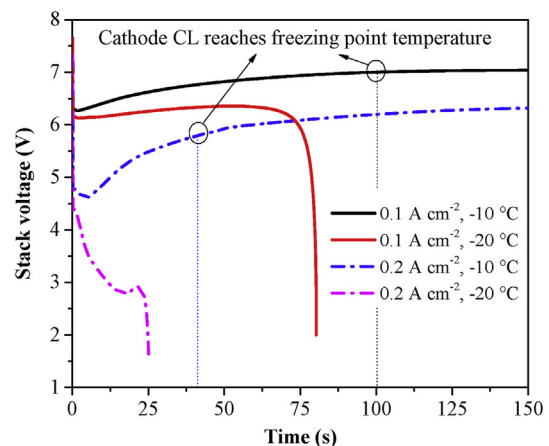


Fig. 3. Stack output voltage under unassisted startup from $-10/-20\text{ }^{\circ}\text{C}$ with the constant current density of $0.1/0.2\text{ A cm}^{-2}$.

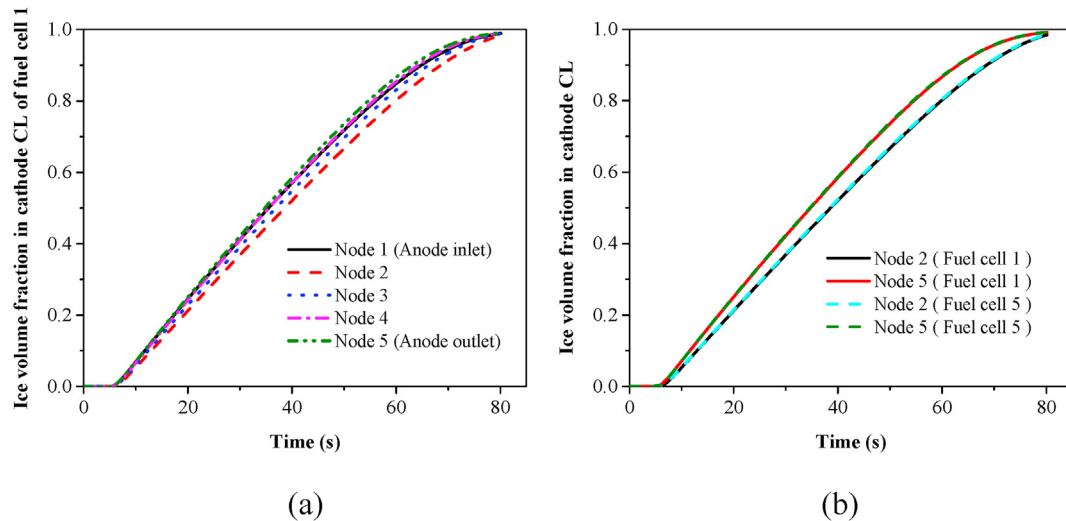


Fig. 4. The distribution of ice volume fraction in cathode CL under unassisted startup from $-20\text{ }^{\circ}\text{C}$, 0.1 A cm^{-2} . (a) In fuel cell 1. (b) In fuel cell 1 and 5.

in the channel direction. As shown in Fig. 4(b), no apparent differences are observed between cell 1 and cell 5, demonstrating the same ice formation rates from the highest (in node 5) to the lowest (in node 2). Under the well heat insulation condition between stack and surroundings ($h = 2\text{ W m}^{-2}\text{ K}^{-1}$ in the presented cases), it is further inferred that there exist no remarkable output voltage variations and temperature variations among individual fuel cells during self-startup processes.

3.3. Reactant gas heating method

Since the stack is not able to successfully start from $-20\text{ }^{\circ}\text{C}$, assisted startup strategies are adopted, including reactant gas heating method, stack heating method, and coolant heating method. Unless otherwise specified, the initial stack temperature is $-20\text{ }^{\circ}\text{C}$, and the current density is 0.1 A cm^{-2} for all assisted startup cases. Fig. 5(a) shows the stack output voltage under different controlling temperatures of humidifier and electrochemical pump, which are arbitrarily kept at $0\text{ }^{\circ}\text{C}$, $20\text{ }^{\circ}\text{C}$, $40\text{ }^{\circ}\text{C}$, and $60\text{ }^{\circ}\text{C}$ during the whole startup process. For the four assisted cases,

startup processes also end in failure although inlet reactant temperatures are significantly increased. In comparison, the assisted cases show slightly higher stack voltage in the first 55 s but the startup processes fail more quickly. With the controlling temperature rising from $0\text{ }^{\circ}\text{C}$ to $60\text{ }^{\circ}\text{C}$, the cold start duration is reduced from about 80 s to about 75 s. It is known that the voltage increase in the early stage of cold start process is typically attributed to the improved membrane hydration, which decreases the ohmic voltage loss [11,44]. Since the current density remains unchanged, the total water generation rate inside stack is considered constant. Therefore, it is inferred that more moisture is introduced into the stack by reactants when the temperature of membrane humidifier and electrochemical hydrogen pump are maintained at higher values. Note that a special case is presented in Fig. 5(a), which only involves the stack model and neglects auxiliary models. The inlet reactant temperature is set as $60\text{ }^{\circ}\text{C}$, and the relative humidity is set as 0. It is found that the cold start duration is about 10 s longer than that of the aforementioned $60\text{ }^{\circ}\text{C}$ assisted case, and the deviation is calculated to be around 13.5%. Therefore, it is concluded that investigating the reactant gas heating method at the stack level

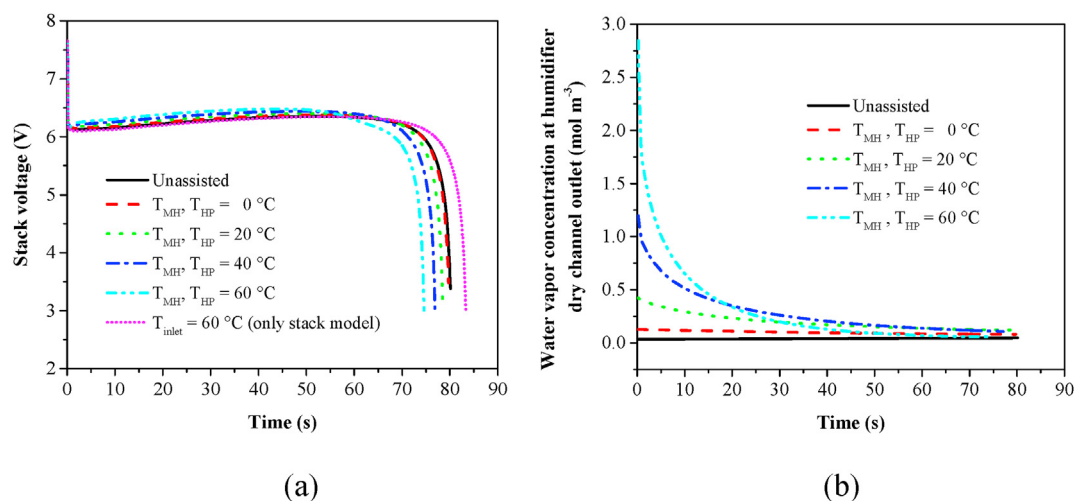


Fig. 5. Effects of reactant gas heating method on cold start performances, $-20\text{ }^{\circ}\text{C}$, 0.1 A cm^{-2} . (a) Stack voltage. (b) Water vapor concentration at membrane humidifier dry channel outlet.

overestimates the cold start survivability because it is unable to consider the disadvantageous moisture coupled with hot reactant gases at the authentic system level.

The corresponding water vapor concentration at the outlet of humidifier dry channel is presented in Fig. 5(b). It is observed that the vapor concentration is around 0.1 mol m^{-3} in the beginning of 0°C case while it reaches about 2.8 mol m^{-3} in the case of 60°C . The differences are caused by the enhanced water transport inside membrane humidifier at higher operating temperatures. With the rise of temperature, the saturated vapor pressure increases exponentially [13]. Consequently, the water activity at humidifier dry side decreases, leading to lower equilibrium membrane water content. As a result, more membrane water turns into water vapor, which subsequently flows out of the dry channel and enters the stack cathode channel. In the humidifier wet channel, the phase change from membrane water to vapor also exists owing to the great temperature difference between humidifier and stack cathode exhaust. Therefore, the membrane water content in humidifier keeps decreasing, resulting in the continuous drop of water vapor concentration as shown in Fig. 5(b). After around 40 s, the water vapor concentration in all cases are gradually stabilized at about 0.2 mol m^{-3} . Based on the accelerated cold start failure, it is found that the advantages of reactant heating effect are not able to compensate for the disadvantages of increased stack inlet relative humidity when started from -20°C . To further reveal the differences of ice generation, the ice volume fraction in cathode CL of fuel cell 1 along the gas channel direction is presented in Fig. 6. The node 1 and node 5 are selected because they represent the inlet and outlet region. Since the variations of ice formation in node 2, node 3, and node 4 are found to be tiny, only the node 3 is shown. With the rise of controlled temperature, the ice generation rate increases obviously in node 1 and node 5 while it remains almost unchanged in node 3. In other words, the inlet regions are strongly affected by the adverse reactant relative humidity while the middle regions suffer minor disadvantageous influences.

The comparison of temperature distribution inside whole stack under the unassisted startup and the reactant gas heating method is presented in Fig. 7. It should be mentioned that stack dimensions in the figure are not to scale as real structural sizes. Several cold start moments are selected, which represent the whole startup duration. For the unassisted case, temperature variations among different fuel cells and along the channel direction are marginal. In a single cell, the temperature of membrane electrode assembly is observed to be slightly higher than other layers because the heat is generated through hydrogen-oxygen reactions in catalyst layers. By maintaining the humidifier and the electrochemical pump at high operating temperatures, more heat is carried by reactants into the stack. Take the 60°C case for an example, the additional heat source along with heated and humidified air is calculated to be around 11.5 W, which is expected to average about 1.15 W for a single fuel cell. For the hydrogen supply, the temperature of recycled hydrogen and water vapor is identical to that of electrochemical pump while the temperature of pure hydrogen from high-pressure tank is the same as surrounding temperature. Since the requested anode stoichiometry ratio is set as 1.2 (listed in Table 1), the hot and wet hydrogen by recirculation only occupies a small portion, leading to the fact that the additional heat carried by anode inlet gases is marginal. As shown in Fig. 7, temperature differences between cathode inlet regions and other regions keep increasing from the beginning to the end. The difference is calculated to be around 1.5 K at 10 s, 2.3 K at 20 s, and 3.0 K at 40 s. By the end of startup process, the temperature difference reaches as high as 4.4 K. Despite obvious temperature variations in the channel direction, the temperature distribution among all fuel cells is relatively uniform. In summary, the effectiveness of reactant heating method is found unsatisfactory because the additional heat carried by gases is averaged to be extremely small for each individual cell. Besides, a large amount of moisture is simultaneously introduced to the stack owing to the increased temperature of membrane humidifier and electrochemical hydrogen pump. Although the stack inlet regions

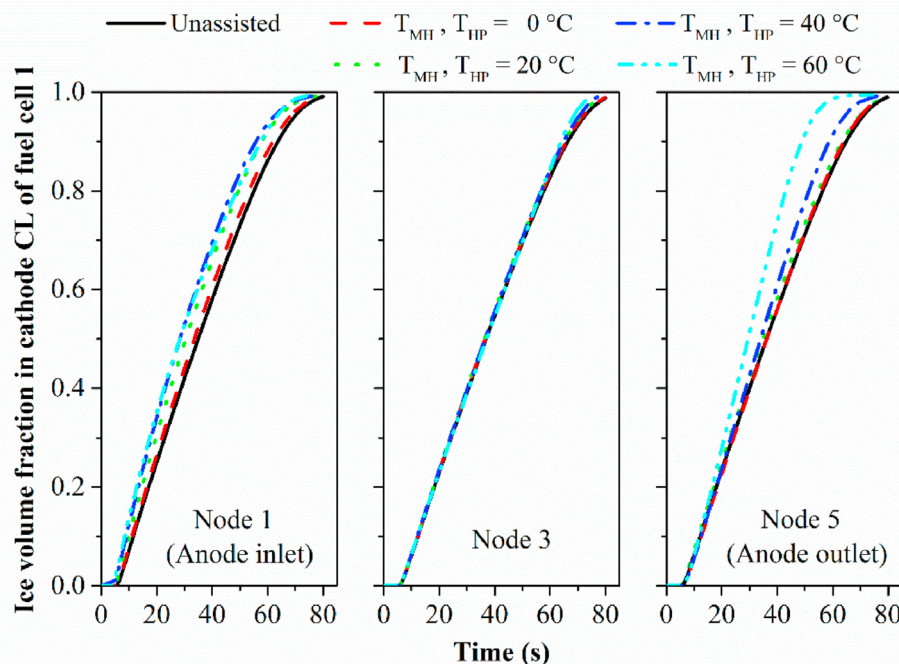


Fig. 6. Ice volume fraction in cathode CL of fuel cell 1 along the gas channel direction, -20°C , 0.1 A cm^{-2} .

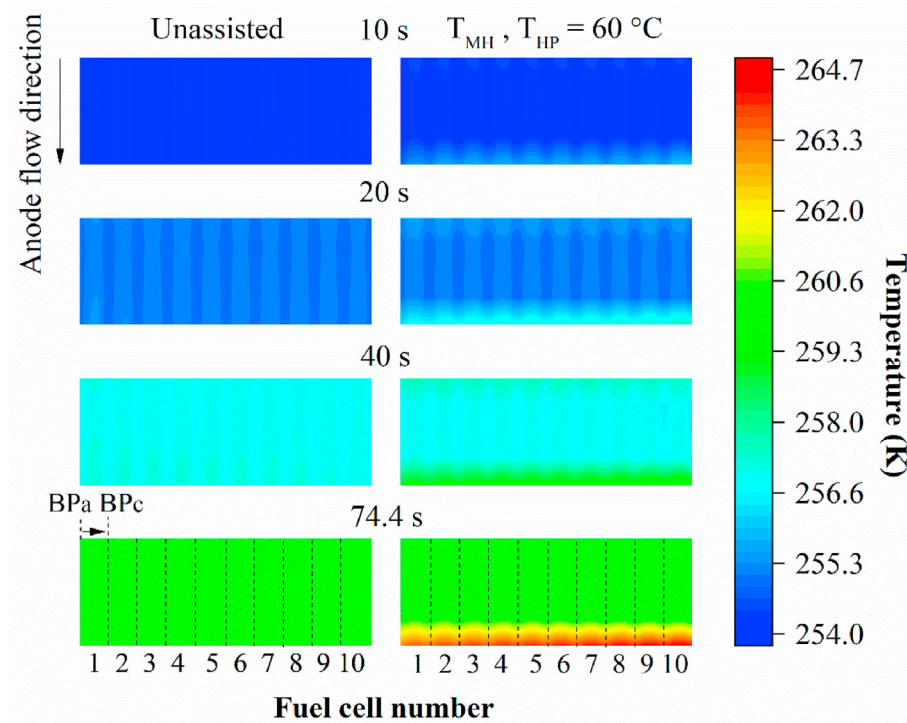


Fig. 7. Temperature distribution inside the whole stack under the unassisted startup and the reactant gas heating method, $-20\text{ }^{\circ}\text{C}$, 0.1 A cm^{-2} .

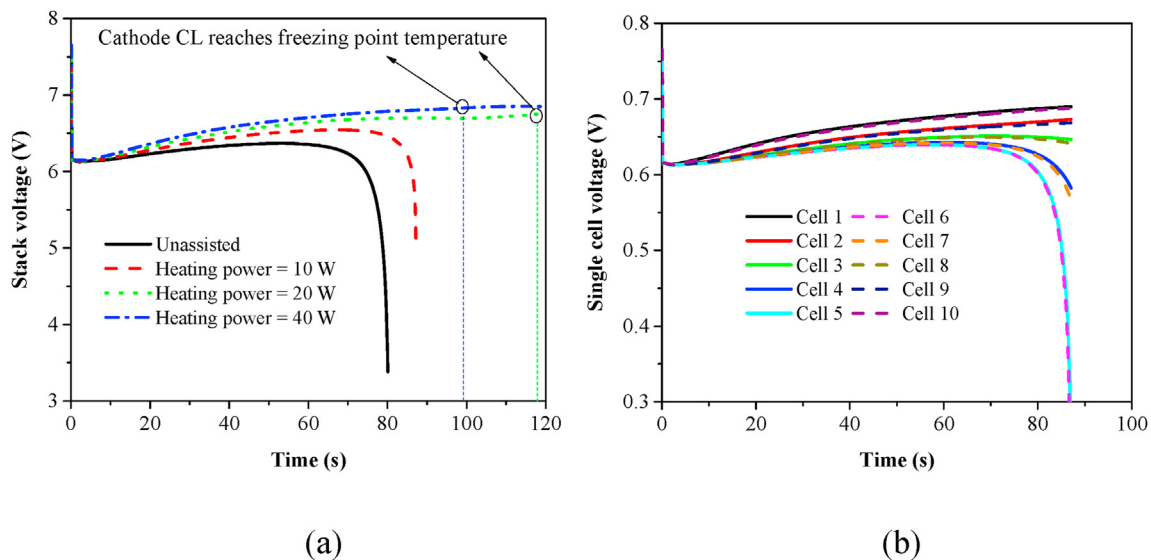


Fig. 8. Effects of different heating powers on cold start performances, $-20\text{ }^{\circ}\text{C}$, 0.1 A cm^{-2} . (a) Stack voltage. (b) Single cell voltage distribution with 10 W heating power.

benefit from high reactant temperatures, they severely suffer from high reactant relative humidity, which may lead to expedited cold start failure.

3.4. Stack heating method

The stack voltage under different heating powers is shown in Fig. 8(a). It should be mentioned that the value of heating power in the figure represents the applied power for one end plate. The total stack heating power is twice as large as the above value. It is observed that the cold start process still fails with 10 W heating

power while it succeeds if the heating power exceeds 20 W . The time when the lowest local temperature of cathode CL inside the whole stack reaches the freezing point temperature is around 117.3 s under 20 W heating power, and it is reduced to approximately 98.5 s at 40 W . The single cell voltage distribution with 10 W heating power is further shown in Fig. 8(b). Cell 1 and cell 10 show the highest voltage while cell 5 and cell 6 demonstrate the lowest output voltage. A symmetrical pattern is observed, indicating that the heating power is indeed applied to the two sides of stack. At the end of startup process, the voltage of cell 5 and cell 6 is reduced to below 0.3 V while the voltage of cell 4 and cell 7 is reduced to about

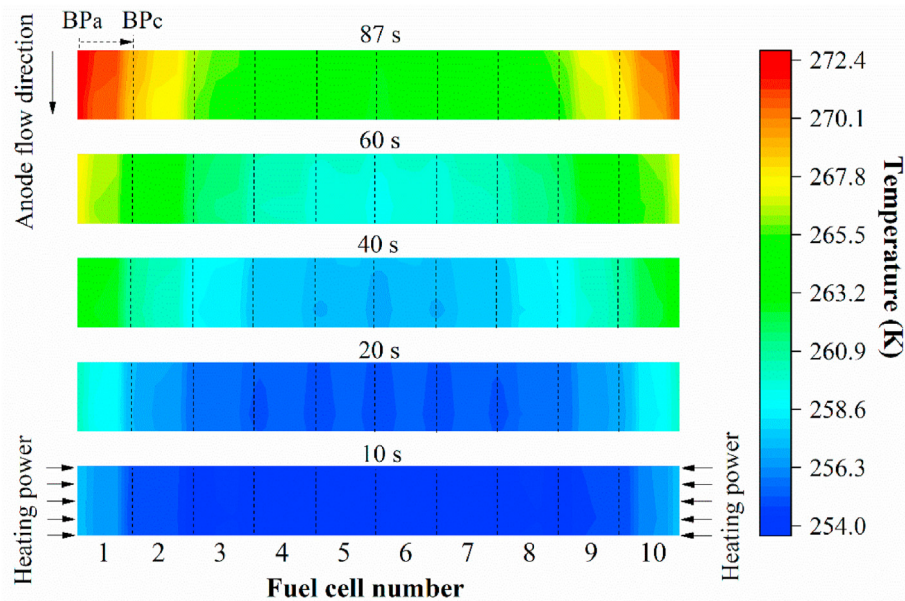


Fig. 9. Temperature distribution among different fuel cells with 10 W heating power, $-20\text{ }^{\circ}\text{C}$, 0.1 A cm^{-2} .

0.58 V. However, the voltage of cell 1, cell 2, cell 3, cell 8, cell 9, and cell 10 does not show obvious decrease. Therefore, it is inferred that the stack shutdown is caused by the startup failure of middle fuel cells while other cells are still operating normally. Under the condition, the variable heating and load control strategy previously proposed by our research group [60] may be applied.

To reveal the detailed parameter distribution throughout the whole startup process, Fig. 9 shows the temperature variations among different fuel cells under 10 W heating power. Since the heating power is applied to end plates, the temperature of end cells is obviously higher than middle cells. However, no remarkable temperature variations are observed in the gas flow direction inside a single cell. At 10 s, the temperature difference between end cells and middle cells is around 3.2 K. The difference grows to 4.8 K at 20 s, 7.1 K at 40 s, 8.2 K at 60 s, and 9.8 K at the end of simulation.

Note that electrochemical reactions are enhanced at higher operating temperatures, leading to the results that output performances of end cells are better than that of middle cells and the corresponding voltage distinctions keep increasing until the cold start failure as shown in Fig. 8(b).

The distribution of ice volume fraction in cathode CL at node 3 with 10 W heating power is shown in Fig. 10. It is found that the ice volume fraction also follows an almost symmetrical pattern throughout the cold start process. For cell 1, the ice fraction reaches 0.03 at 10 s, 0.12 at 20 s, 0.22 at 40 s, 0.28 at 60 s, and 0.29 at 87 s. For cell 5, the ice volume fraction increases to 0.06 at 10 s, 0.23 at 20 s, 0.54 at 40 s, 0.80 at 60 s, and 0.99 at 87 s. It is obvious that cell 1, cell 2, cell 9, and cell 10 are still far away from being fully blocked by ice at the end of simulation. Therefore, the successful startup of middle cells can be regarded as a crucial index for the successful cold start of whole stack. Table 6 lists the maximum ice volume fraction in cathode CL, maximum temperature difference, and cold start duration with different heating powers ranging from 0 W to 60 W. The maximum temperature difference is calculated between two componential layers within the entire stack, and the cold start duration is counted as the exact time when the lowest CL temperature reaches the freezing point temperature. With 20 W heating power, the startup duration is about 117.3 s, and the maximum temperature difference is about 22.0 K. By increasing the power from 20 W to 30 W, the cold start duration is reduced by 10.7 s. Meanwhile, the startup duration is decreased by 8.1 s from 30 W to 40 W, 6.2 s from 40 W to 50 W, and 5.4 s from 50 W to 60 W, indicating that the degree of time reduction with every 10 W additional heating power is decreasing. Note that the maximum temperature difference is simultaneously increased to 33.6 K at 30 W, 45.9 K at 40 W, 56.6 K at 50 W, and 67.7 K at 60 W. It is inferred that the cold start duration will not decrease anymore if the heating power reaches certain values, but the maximum temperature difference will keep rising if the heating power is further increased. The reason is that insufficient thermal conductivities of fuel cell componential layers make it hard to rapidly transport the heat from end plates to middle fuel cells. Therefore, improving the thermal conductivity is of great importance to further accelerate the cold start process. Besides, the stack heating power should be

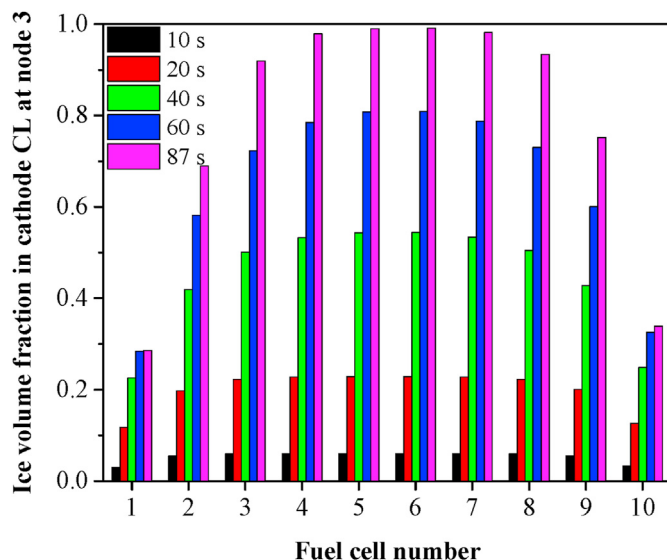


Fig. 10. Distribution of ice volume fraction in cathode CL at node 3 with 10 W heating power, $-20\text{ }^{\circ}\text{C}$, 0.1 A cm^{-2} .

Table 6

Maximum ice volume fraction in cathode CL, maximum temperature difference, and cold start duration with different heating powers.

Heating power (W)	Maximum ice volume fraction in cathode CL	Maximum temperature difference (K)	Cold start duration (s)
0 (Unassisted)	0.992	0.7	80.1 (failed)
10	0.993	9.8	87.0 (failed)
15	0.992	14.9	93.3 (failed)
20	0.971	22.0	117.3
25	0.942	27.7	112.2
30	0.887	33.6	106.6
40	0.789	45.9	98.5
50	0.718	56.6	92.3
60	0.664	67.7	86.9

carefully selected because large temperature gradients among the stack result in disadvantageous thermal stresses while large ice volume fraction inside porous layers is likely to cause irreversible structural damages, which are both harmful for the stack durability.

3.5. Coolant heating method

Since coolant is able to absorb heat from the stack under normal operating temperatures, it is also possible for coolant to heat the stack under cold start conditions. By placing a heater in the heat management subsystem, the coolant temperature can be largely increased before flowing into the stack. Fig. 11 shows the stack voltage under different coolant inlet temperatures. It is found that the stack voltage improves with higher coolant temperatures, which results from the enhanced fuel cell operating temperature. Although the stack still fails when the coolant temperature is -10°C , the cold start process succeeds when the temperature reaches -5°C . In addition, the startup duration is reduced to about 72.8 s if the coolant temperature further rises to 0°C . Table 7 lists the maximum ice volume fraction in cathode CL, maximum temperature difference, and cold start duration with different coolant inlet temperatures from -10°C to 20°C . Small temperature differences are observed, indicating that the temperature distribution is relatively uniform inside the whole stack. Unlike the stack heating method, the additional heat carried by coolant is evenly distributed into each fuel cells, which raises the stack temperature uniformly and rapidly. When the coolant inlet temperature is kept at 10°C , the ice formation can even be avoided, which is of great

importance for cold start strategy optimizations. However, it should be noted that the power consumption for heating coolant is extremely large. Take the coolant composed of 50% glycol and 50% water for an example, the power requirement to raise 0.1 kg s^{-1} coolant from -20°C to 0°C is calculated to be around 6.2 kW. Therefore, it is unrealistic to merely rely on the stack output power to satisfy the power consumption, implying that secondary power sources are necessary for the coolant heating method.

The ice volume fraction in cathode CL of fuel cell 1 is shown in Fig. 12(a), and the corresponding CL temperature is shown in Fig. 12(b). The node 3 is selected because it represents the center region of fuel cell in channel direction. It is observed that the ice generation rate changes in the case of -10°C since the curve slope is not constant. Before around 60 s, the slope keeps decreasing slightly, and it is smaller than that of the unassisted case. Subsequently, the slope increases insignificantly, which indicates that the ice formation grows faster. In the case of -5°C , the curve slope also changes throughout the whole process. As shown in Fig. 12(b), the CL temperature indeed increases much faster with the coolant heating method in comparison with the unassisted case. Note that the temperature difference between coolant and stack is large in the beginning, which correspondingly results in the high temperature increase rate. With the CL temperature rising, the curve slope drops, indicating a decreased temperature growing rate. As stated in the phase change mechanisms, the super-cooled water is generated if the membrane water gets saturated while the saturated membrane water content is dependent on the local temperature. The heat absorbed from the coolant accelerates the stack temperature increment, which decreases the difference between the membrane water content and the saturated content, leading to lower ice formation rates. Furthermore, if the membrane hydration rate through electrochemical reactions is hard to catch the growing rate of saturated membrane water content, the ice formation will be strongly suppressed.

4. Conclusions

In the study, an integrated proton exchange membrane fuel cell system model, which is composed of stack, membrane humidifier, electrochemical hydrogen pump, compressor, and radiator, is developed to investigate the cold start performances. The integration of all sub-models has been presented in detail. All sub-models have been validated against experimental data, especially the pseudo two-dimensional stack sub-model. The unassisted cold start performances are firstly demonstrated. The stack is able to reach the freezing point temperature when started from -10°C while it fails when started from -20°C . It is further observed that the ice volume fraction of cathode catalyst layer varies in the gas channel direction inside a single fuel cell. However, there exist no remarkable output voltage variations and temperature variations among individual fuel cells.

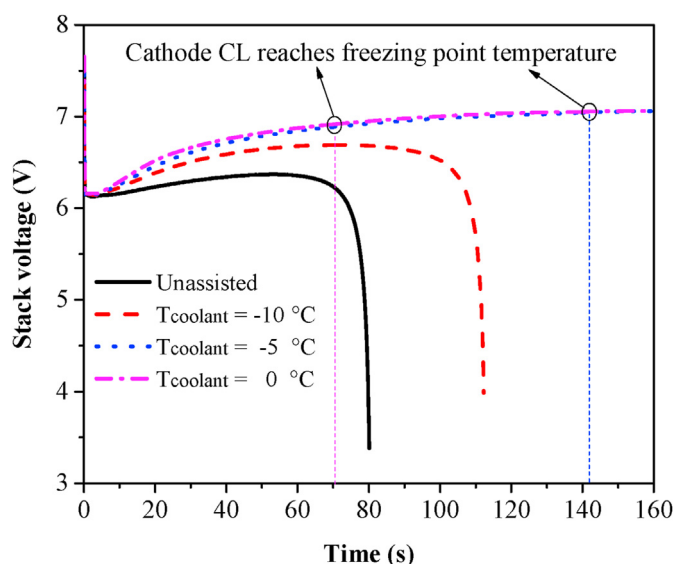
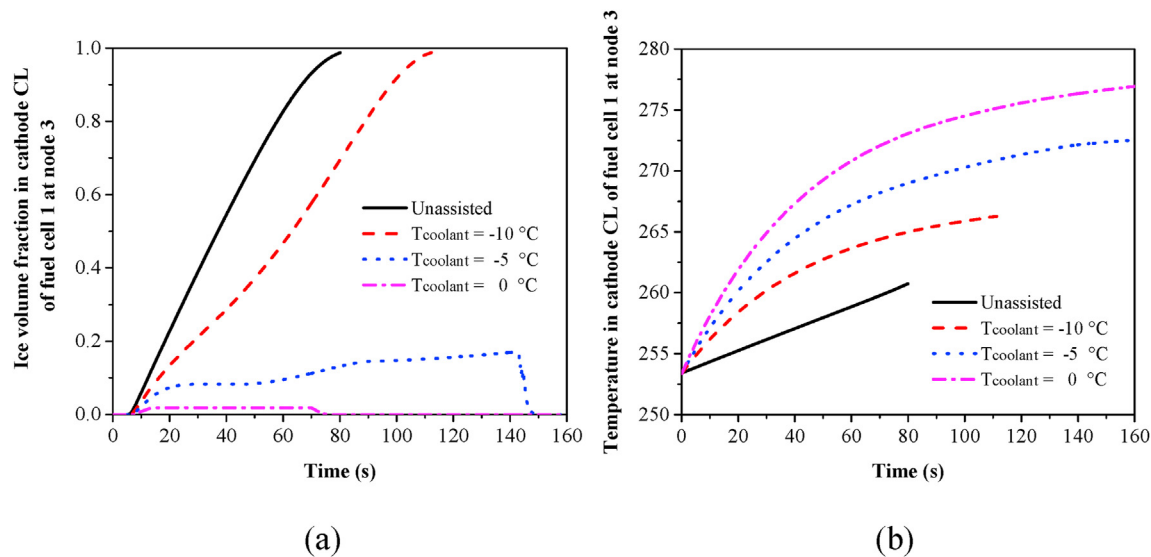


Fig. 11. Stack voltage under different coolant temperatures, -20°C , 0.1 A cm^{-2} .

Table 7

Maximum ice volume fraction in cathode CL, maximum temperature difference, and cold start duration with different coolant inlet temperatures.

Coolant inlet temperature (°C)	Maximum ice volume fraction in cathode CL	Maximum temperature difference (K)	Cold start duration (s)
−10	0.992	0.8	112.3 (failed)
−5	0.266	1.3	143.2
0	0.036	0.8	72.8
5	0.006	0.6	48.4
10	<0.001	0.5	36.9
20	<0.001	0.4	25.6

**Fig. 12.** Cold start performances of fuel cell 1 under different coolant temperatures, $-20\text{ }^{\circ}\text{C}$, 0.1 A cm^{-2} . (a) Ice volume fraction in cathode CL at node 3. (b) Temperature of cathode CL at node 3.

To successfully start from $-20\text{ }^{\circ}\text{C}$, various assisted startup strategies are subsequently investigated. The stack voltage, single cell voltage distribution, stack inlet relative humidity, ice volume fraction, temperature distribution, and cold start duration are comprehensively compared. Although the temperature of membrane humidifier and electrochemical hydrogen pump are maintained at $60\text{ }^{\circ}\text{C}$, the amount of additional heat carried by gases is averaged less than 1.2 W for each individual fuel cell. Note that a large amount of moisture is simultaneously introduced to the stack. It is found that the advantages of reactant heating effect are not able to compensate for the disadvantages of increased stack inlet relative humidity, which may lead to expedited cold start failure. For stack heating method, the cold start process succeeds if the total heating power exceeds 40 W . Since the output performances of end fuel cells are better than middle cells, the successful startup of middle cells can be regarded as a crucial index for the whole stack. With heating power increasing from 20 W to 60 W , the cold start duration is reduced from 117.3 s to 86.9 s . However, the corresponding maximum temperature difference within whole stack is increased from 22.0 K to 67.7 K . Therefore, improving the thermal conductivity of fuel cell materials is of great importance. For coolant heating method, the cold start succeeds when the coolant temperature reaches $-5\text{ }^{\circ}\text{C}$. The ice formation can even be avoided if the coolant inlet temperature is kept at $10\text{ }^{\circ}\text{C}$. In addition, the temperature distribution is relatively uniform inside the whole stack. However, the power consumption for heating coolant is extremely large, and it is unrealistic to merely rely on the stack output power to satisfy the power requirement, implying that secondary power sources are necessary.

Author statement

Zirong Yang: Methodology, Software, Validation, Writing – original draft, Kui Jiao: Conceptualization, Resources, Writing – review & editing, Kangcheng Wu: Validation, Writing – review & editing, Weilong Shi: Writing – review & editing, Funding acquisition. Shangfeng Jiang: Writing – review & editing, Funding acquisition. Longhai Zhang: Writing – review & editing, Funding acquisition. Qing Du: Writing – review & editing, Supervision, Project administration

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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